

Riemann zeta recurrence relation

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Abstract

The Riemann zeta function can be written as a recurrence relation. For $\Re(s) > 1$:

$$\zeta(s) = 1 + \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{1}{p_n^s} \cdot S_{p_{n-1}}(0, m, s)$$

Where p_n is the n -th prime, and $S_{p_n}(a, m, s)$ is the recurrence relation:

$$S_{p_0}(a, m, s) = \frac{1}{(1 + a + m)^s}$$
$$S_{p_n}(a, m, s) = \left(\sum_{k=0}^{p_n-1} S_{p_{n-1}}(a + k \cdot p_{n-1}\#, p_n \cdot m, s) \right) - \frac{1}{p_n^s} \cdot S_{p_{n-1}}\left(\frac{a}{p_n}, m, s\right)$$

Where p_0 is the zeroth prime, and $p_n\#$ is the n -th primorial.

Riemann zeta recurrence relation

Consider $\zeta(s)$ in Dirichlet series form, convergent for $\Re(s) > 1$.

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$
$$= 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{4^s} + \frac{1}{5^s} + \frac{1}{6^s} + \frac{1}{7^s} + \dots$$

Group the terms by least prime factor.

$$\begin{aligned}
\zeta(s) &= 1 + \frac{1}{2^s} + \frac{1}{3^s} + \frac{1}{(2 \cdot 2)^s} + \frac{1}{5^s} + \frac{1}{(2 \cdot 3)^s} + \frac{1}{7^s} + \dots \\
&= 1 + \left(\frac{1}{2^s} + \frac{1}{(2 \cdot 2)^s} + \frac{1}{(2 \cdot 3)^s} + \dots \right) + \left(\frac{1}{3^s} + \frac{1}{(3 \cdot 3)^s} + \frac{1}{(3 \cdot 5)^s} + \dots \right) + \dots
\end{aligned}$$

Group by cyclic patterns of numbers coprime with the primorial.

$$\begin{aligned}
\zeta(s) &= 1 + \sum_{m=0}^{\infty} \frac{1}{(2 + 2 \cdot m)^s} + \sum_{m=0}^{\infty} \frac{1}{(3 + 6 \cdot m)^s} + \sum_{m=0}^{\infty} \frac{1}{(5 + 30 \cdot m)^s} + \sum_{m=0}^{\infty} \frac{1}{(25 + 30 \cdot m)^s} + \dots \\
&= 1 + \sum_{m=0}^{\infty} \left(\frac{1}{2^s} \cdot \frac{1}{(1+m)^s} + \frac{1}{3^s} \cdot \frac{1}{(1+2 \cdot m)^s} + \frac{1}{5^s} \cdot \left(\frac{1}{(1+6 \cdot m)^s} + \frac{1}{(5+6 \cdot m)^s} \right) + \dots \right)
\end{aligned}$$

The primorial patterns in the terms can be written as a recurrence relation.

For $\Re(s) > 1$:

$$\zeta(s) = 1 + \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} \frac{1}{p_n^s} \cdot S_{p_{n-1}}(0, m, s)$$

Where p_n is the n -th prime, and $S_{p_n}(a, m, s)$ is the recurrence relation:

$$\begin{aligned}
S_{p_0}(a, m, s) &= \frac{1}{(1 + a + m)^s} \\
S_2(a, m, s) &= \frac{1}{(1 + a + 2 \cdot m)^s} \\
S_3(a, m, s) &= \frac{1}{(1 + a + 6 \cdot m)^s} + \frac{1}{(5 + a + 6 \cdot m)^s} \\
S_{p_n}(a, m, s) &= \left(\sum_{k=0}^{p_n-1} S_{p_{n-1}}(a + k \cdot p_{n-1}\#, p_n \cdot m, s) \right) - \frac{1}{p_n^s} \cdot S_{p_{n-1}}\left(\frac{a}{p_n}, m, s\right)
\end{aligned}$$

Where p_0 is the zeroth prime (initial condition), and $p_n\#$ is the n -th primorial.

The positive terms correspond with the numbers coprime with the primorial, or *candidate primes*. The negative part is the candidate prime eliminator. The eliminator is the pattern of the prior prime in the sequence of least prime factors, stretched by the new prime.

By negating the signs of the even terms we obtain Dirichlet eta $\eta(s)$, which is convergent for $\Re(s) > 0$. Dirichlet $\eta(s)$ has the same non-trivial zeros as $\zeta(s)$. All the even terms are covered by $n = 1$, so we can write:

$$\begin{aligned}\eta(s) &= 1 + \sum_{m=0}^{\infty} \left(-\frac{1}{(2+2 \cdot m)^s} + \sum_{n=2}^{\infty} \frac{1}{p_n^s} \cdot S_{p_{n-1}}(0, m, s) \right) \\ &= 1 - \sum_{m=1}^{\infty} \frac{1}{(2 \cdot m)^s} + \sum_{m=0}^{\infty} \sum_{n=2}^{\infty} \frac{1}{p_n^s} \cdot S_{p_{n-1}}(0, m, s) \\ &= 1 - \frac{1}{2^s} \cdot \zeta(s) + \sum_{m=0}^{\infty} \sum_{n=2}^{\infty} \frac{1}{p_n^s} \cdot S_{p_{n-1}}(0, m, s)\end{aligned}$$

How to apply the recurrence relation

Applying the recurrence relation to obtain S_2 , S_3 and S_5 .

$$\begin{aligned}S_{p_0}(a, m, s) &= \frac{1}{(1+a+m)^s} \\ S_{p_n}(a, m, s) &= \left(\sum_{k=0}^{p_n-1} S_{p_{n-1}}(a+k \cdot p_{n-1}\#, p_n \cdot m, s) \right) - \frac{1}{p_n^s} \cdot S_{p_{n-1}}\left(\frac{a}{p_n}, m, s\right)\end{aligned}$$

Solving for S_2 :

$$\begin{aligned}S_2(a, m, s) &= S_{p_0}(a, 2 \cdot m) + S_{p_0}(a+p_0\#, 2 \cdot m, s) - \frac{1}{2^s} \cdot S_{p_0}\left(\frac{a}{2}, m, s\right) \\ &= \frac{1}{(1+a+2 \cdot m)^s} + \frac{1}{(2+a+2 \cdot m)^s} - \frac{1}{(2+a+2 \cdot m)^s} \\ S_2(a, m, s) &= \frac{1}{(1+a+2 \cdot m)^s}\end{aligned}$$

Solving for S_3 :

$$\begin{aligned}S_3(a, m, s) &= \left(\sum_{k=0}^{3-1} S_2(a+k \cdot 2\#, 3 \cdot m, s) \right) - \frac{1}{3^s} \cdot S_2\left(\frac{a}{3}, m, s\right) \\ &= S_2(a, m \cdot 3, s) + S_2(a+2, 3 \cdot m, s) + S_2(a+4, 3 \cdot m, s) - \frac{1}{(3+a+6 \cdot m)^s} \\ &= \frac{1}{(1+a+6 \cdot m)^s} + \frac{1}{(3+a+6 \cdot m)^s} + \frac{1}{(5+a+6 \cdot m)^s} - \frac{1}{(3+a+6 \cdot m)^s} \\ S_3(a, m, s) &= \frac{1}{(1+a+6 \cdot m)^s} + \frac{1}{(5+a+6 \cdot m)^s}\end{aligned}$$

Solving for S_5 :

$$\begin{aligned}
S_5(a, m, s) &= \left(\sum_{k=0}^{5-1} S_3(a + k \cdot 3\#, 5 \cdot m, s) \right) - \frac{1}{5^s} \cdot S_3\left(\frac{a}{5}, m, s\right) \\
&= S_3(a, 5 \cdot m, s) + S_3(a + 6, 5 \cdot m, s) + S_3(a + 12, 5 \cdot m, s) \\
&\quad + S_3(a + 18, m \cdot 5, s) + S_3(a + 24, 5 \cdot m, s) - \frac{1}{5^s} \cdot S_3\left(\frac{a}{5}, m, s\right) \\
&= \frac{1}{(1 + a + 30 \cdot m)^s} + \frac{1}{(5 + a + 30 \cdot m)^s} \\
&\quad + \frac{1}{(7 + a + 30 \cdot m)^s} + \frac{1}{(11 + a + 30 \cdot m)^s} \\
&\quad + \frac{1}{(13 + a + 30 \cdot m)^s} + \frac{1}{(17 + a + 30 \cdot m)^s} \\
&\quad + \frac{1}{(19 + a + 30 \cdot m)^s} + \frac{1}{(23 + a + 30 \cdot m)^s} \\
&\quad + \frac{1}{(25 + a + 30 \cdot m)^s} + \frac{1}{(29 + a + 30 \cdot m)^s} \\
&\quad - \frac{1}{(5 + a + 30 \cdot m)^s} - \frac{1}{(25 + a + 30 \cdot m)^s} \\
S_5(a, m, s) &= \frac{1}{(1 + a + 30 \cdot m)^s} + \frac{1}{(7 + a + 30 \cdot m)^s} + \frac{1}{(11 + a + 30 \cdot m)^s} \\
&\quad + \frac{1}{(13 + a + 30 \cdot m)^s} + \frac{1}{(17 + a + 30 \cdot m)^s} + \frac{1}{(19 + a + 30 \cdot m)^s} \\
&\quad + \frac{1}{(23 + a + 30 \cdot m)^s} + \frac{1}{(29 + a + 30 \cdot m)^s}
\end{aligned}$$

Congruence relations and residue systems

As n increases, the growth in the number of terms in S_{p_n} is the same as the growth in the number of congruence relations.

Any prime p greater than 2 satisfies the congruence relation:

$$p \equiv 1 \pmod{2}$$

Any prime p greater than 3 satisfies one of the congruence relations:

$$p \equiv 1 \pmod{6}$$

$$p \equiv 5 \pmod{6}$$

Any prime p greater than 5 satisfies one of the congruence relations:

$$\begin{aligned}
p &\equiv 1 \pmod{30} \\
p &\equiv 7 \pmod{30} \\
p &\equiv 11 \pmod{30} \\
p &\equiv 13 \pmod{30} \\
p &\equiv 17 \pmod{30} \\
p &\equiv 19 \pmod{30} \\
p &\equiv 23 \pmod{30} \\
p &\equiv 29 \pmod{30}
\end{aligned}$$

The number of congruence relations C_{p_n} added at step n is equal to the number of coprimes with the n -th primorial:

$$\begin{aligned}
C_{p_n} &= \prod_{i=1}^n (p_i - 1) \\
&= \phi(p_n\#)
\end{aligned}$$

Where $p_n\#$ is the n -th primorial, and ϕ is Euler's totient function.